

CODE:

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
import os

img = cv2.imread("/content/Gemini_Generated_Image_tx2udatx2udatx2u.png")
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

H, W = gray.shape
factors = [2, 4, 8, 16]

os.makedirs("output", exist_ok=True)

cv2.imwrite("output/original_gray.jpg", gray)

for k in factors:
    new_h = H // k
    new_w = W // k
    down = cv2.resize(gray, (new_w, new_h), interpolation=cv2.INTER_AREA)
    up = cv2.resize(down, (W, H), interpolation=cv2.INTER_NEAREST)
    cv2.imwrite(f"output/spatial_1_{k}.jpg", up)

F = np.fft.fft2(gray)
F_shift = np.fft.fftshift(F)

for k in factors:
    keep_h = H // k
    keep_w = W // k
    mask = np.zeros((H, W), np.uint8)
    r1 = H//2 - keep_h//2
    r2 = r1 + keep_h
    c1 = W//2 - keep_w//2
    c2 = c1 + keep_w
    mask[r1:r2, c1:c2] = 1
    F_masked = F_shift * mask
    mag = 20 * np.log(np.abs(F_masked) + 1)
    mag_norm = cv2.normalize(mag, None, 0, 255, cv2.NORM_MINMAX).astype(np.uint8)
    cv2.imwrite(f"output/spectrum_1_{k}.jpg", mag_norm)
    F_ishift = np.fft.ifftshift(F_masked)
    img_back = np.fft.ifft2(F_ishift)
    img_back = np.real(img_back)
    img_back_norm = cv2.normalize(img_back, None, 0, 255, cv2.NORM_MINMAX).astype(np.uint8)
    cv2.imwrite(f"output/frequency_1_{k}.jpg", img_back_norm)

print("All outputs saved in the 'output' folder.")
```